

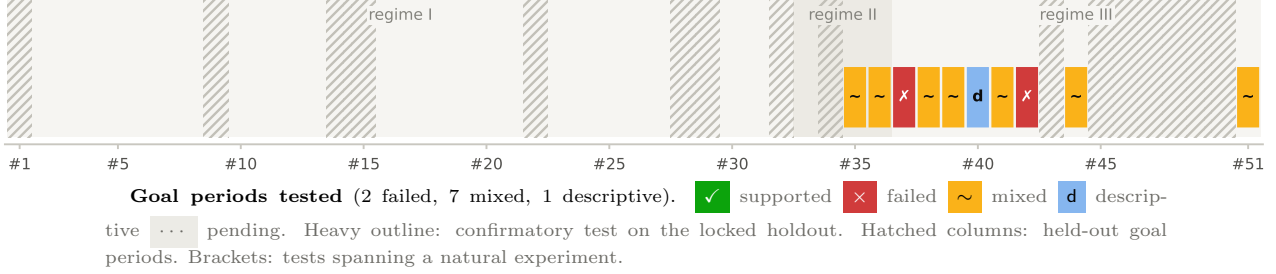
H26: The swarm is near-critical in what it says but subcritical in when it acts

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 1/5 Usefulness 2/5

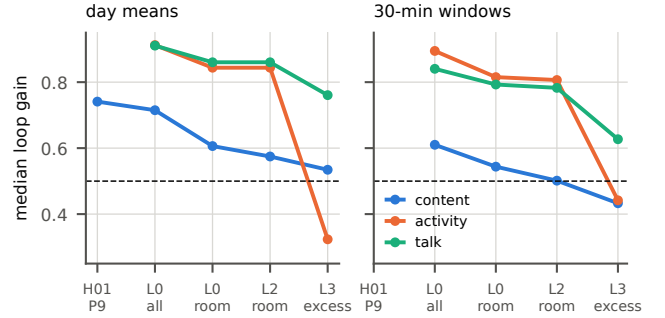
Bottom line. At matched time resolution and drive removal, LLM-swarm content and activity have similar room loop gains (≈ 0.5); H01's 0.74 is mostly drives plus time-averaging, and the channels differ in *where* they co-move, not how strongly.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

HH108 claimed the swarm's *content* is near-critical (H01: $\beta J_0/n = 0.74$) while its *activity* is subcritical (loop gains ≤ 0.4). We put content, activity and talk on one estimand, $g = 1 - 1/VR$, at matched resolution (day means, 30-min windows). Field removal was nested: dedupe, goal and kick-off directions, operator-message directions, and a within-room minus cross-room contrast. We ran 10 regime-III two-room units plus #35 and #51. Synthetic worlds at village sampling came first. Drives alone reproduce H01's number: 0.57 from a global drive at the observed cross-room level, 0.74 with more drive types. Real data: content's room excess is genuine (median 0.53 [0.34, 0.66]; room-permutation null beaten in 6/10 units). But activity matches it at 30 min (0.44 vs 0.43) and is larger in #51. Verdicts: 0 supported, 8 mixed, 2 failed. **Failed as posed.** Holdout script written, not run.



Observables. Median loop gain over 10 regime-III two-room units, down the field-removal ladder: H01's estimator, then whole-swarm (L0 all), room (L0), drive-projected room (L2), and room excess (L3). Content (blue) falls from 0.74 to 0.53 (day) and 0.43 (30 min). Activity and talk sit above content at every step except day-level activity L3, which is unpowered. At 30 min, activity's excess equals content's. Activity's large raw gain is village-wide co-movement.

PHYSICAL MODEL

Mean-field soft-spin $O(n)$ / Curie-Weiss (models 11, 01). Spins: an agent's 32-d whitened statement mean (content), or its fraction of active or talking minutes. Within-room coupling J ; drives: global day and time-of-day, static goal fields, measured operator inputs, unmeasured room drives. C_{ab} : cross-agent covariance of drive-removed deviations; S_a : split-half signal variance; ρ_w, ρ_c : within- and cross-room correlation; N_r : room size.

$$\rho_{w,c} = \frac{\sum_{\text{pairs}} \text{tr} C_{ab}}{\sum_{\text{pairs}} \sqrt{S_a S_b}}, \quad \rho_{\text{ex}} = \frac{\rho_w - \rho_c}{1 - \rho_c}$$

$$g = 1 - \frac{1}{1 + (N_r - 1) \rho_{\text{ex}}}, \quad g_{\text{eq}} = J, \quad g_{\text{day}} \rightarrow J(2 - J)$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Mostly a correction, with one usable diagnostic. The HH108 gap was an artifact: it set a day-mean statistic that includes drives against a 1-min statistic with drives removed. Time-averaging alone turns an equal-time gain of 0.4 into 0.64. Compare channels at the same resolution and with the same drive removal and the gap disappears: ideas do not cascade more easily than effort. The diagnostic is *where* co-movement lives. Content co-moves within rooms (cross-room $\rho_c \approx 0.05$). Activity co-moves village-wide (ρ_c 0.5–0.8), so a swarm-wide activity dip is a platform or schedule event. Operator messages explain none of the room content co-movement. Near-critical is too strong for either channel: room loop gains of about 0.5 mean variance about twice the independent level.

H26-content-near-critical · generated 2026-10-03 19:29 by infra/summaries/build_summaries.py · card hypotheses/H26-content-near-critical/README.md · ratings: H26 agent (suggested)

RESULTS VS PREDICTIONS

Prediction (locked)	Observed	Verdict
P1a drives alone give H01 ≥ 0.3	0.57 (global); 0.67–0.74 (more drives)	passed
P1b L3 unbiased, no channel bias	global -0.01 ; content $+0.25$, activity -0.15	partial
P2 L3 content median 0.25–0.5	0.53 [0.34, 0.66]; ≥ 0.5 in 6/10	failed (higher)
P3 content – activity > 0.15	day 6/10 (CI > 0: 1); 30 min 4/10	failed
P3 content – talk > 0.15 (30 min)	2/10; median -0.09	failed
P4 day beyond time-averaging	0.53 vs map 0.68	failed
P5 operator inputs plain $\leq 10\%$	-0.16 to $+0.05$, all units	passed
P6 #51: content $\geq$ activity	reversed in 4/4 (0.25–0.65 vs 0.92–0.95)	failed
Card rule	0 supported, 8 mixed, 2 failed	failed (R2)
Post hoc (H25): N_r^α	$\rho_w \propto$ content -1.4 , activity -0.1	lead

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	0
D unfitted predictions	0	I transfer	0
E interventional	0		

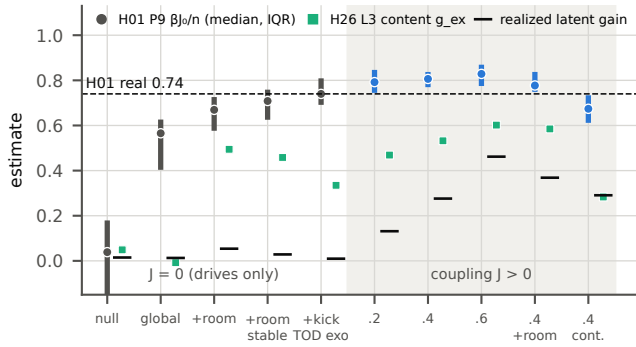
Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Unmeasured room drives (forks in #35, instructions in #38, the override in #44) read as coupling. Day-level activity is nearly unpowered (3–8 days). Talk’s split-half variance is unreliable. The estimator favors content in synthetic worlds, which strengthens the null. One-room units lack a baseline. One embedding model; no cascade sizes (H34). The post hoc size scaling is confounded with #51.

ROUND 2

- **Essence.** whether ideas spread through a swarm more easily than effort does, measured so that channels are comparable. Round 1 says: not at matched resolution; the channels differ in where they co-move, not in how much.
- **H26-R1** Separate room coupling from room drive with the merge and split natural experiments.
- **H26-R2** Lagged, exposure-conditioned content gain.
- **H26-R3** Idea-cascade sizes (HH122) from H34.
- **H26-R4** Instrument checks: embedding swap, style control, H38 outages.



Second figure. Synthetic validation (24 worlds each, $N = 14$ in two rooms, 5 days). With zero coupling, H01’s P9 (gray) already reads 0.57–0.74 from drives and barely moves with J . H26’s L3 (green) tracks the realized latent gain (black) under global drives. It reads unmeasured room drives (“+room”) and imperfectly measured operator pulses as coupling.

GOAL PERIODS IN DETAIL

- G35 ~ mixed: content room excess 0.80 ($N2$ p 0.01), but activity nearly matches it at w30 once its ...
- G36 ~ mixed: content 0.53; activity excess noisy (4 days); at w30 activity exceeds content (gap -0...)
- G37 × failed: content co-fluctuation is global (within-room 0.23, cross-room 0.20); L3 only 0.15; ...
- G38 ~ mixed: content 0.65, 0.55, 0.40 in 38a, b, c ($N2$ p up to 0.03); at w30 activity ties or exce...
- G39 ~ mixed: content 0.45; activity room excess negative (cross-room above within), which inflates...
- G40 d descriptive: rooms merged (not two-room by rule); whole-room gains: content 0.75, activity 0...
- G41 ~ mixed: best case: content 0.77 [0.72, 0.80], $N2$ p 0.005; activity’s whole-swarm co-fluctuati...
- G42 × failed: content co-fluctuation is global (within-room 0.14, cross-room 0.12); L3 0.13; activ...
- G44 ~ mixed: content 0.76; at w30 activity 0.81 and talk 0.78 exceed content (gap -0.09)
- G51 ~ mixed: one big room: activity 0.92 to 0.95 far above content 0.25 to 0.65 (L2 upper bounds);...